FCSG-Net: Frequency-Band Spatial Expert Routing for Lightweight Composite Image Restoration

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National Institute of Technology Rourkela

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FCSG-Net: Frequency-Band Spatial Expert Routing for Lightweight Composite Image Restoration

Project Report submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology

in

[TODO: Department name]

by

[**TODO:** Student Name 1 (Roll No.)]
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under the supervision of

[**TODO:** Prof. Supervisor Name]



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National Institute of Technology Rourkela

November 2026

Certificate of Examination

This is to certify that the work in the Project Report entitled *FCSG-Net: Frequency-Band Spatial Expert Routing for Lightweight Composite Image Restoration* by [**TODO:** Student Name 1 (Roll No.)] [**TODO:** Student Name 2 (Roll No.)] has been examined and is accepted in partial fulfillment of the requirements for the degree of Bachelor of Technology in [**TODO:** Department name].

Supervisor

Examiner

Supervisors' Certificate

This is to certify that the work presented in this Project Report entitled *FCSG-Net: Frequency-Band Spatial Expert Routing for Lightweight Composite Image Restoration* by [**TODO:** Student Name 1 (Roll No.)] [**TODO:** Student Name 2 (Roll No.)] is a record of original research carried out under our supervision and guidance in partial fulfillment of the requirements for the degree of Bachelor of Technology in [**TODO:** Department name]. Neither this Project Report nor any part of it has been submitted for any degree or diploma elsewhere.

[**TODO:** Prof. Supervisor Name]

[**TODO:** Prof. Co-Supervisor Name]

*[**TODO:** Dedication, or remove \dedication from main.tex]*

Declaration of Originality

We hereby declare that this Project Report is our own work and that, to the best of our knowledge, it contains no material previously published or written by another person, nor material which has been accepted for the award of any other degree, except where due acknowledgement is made in the text. Results reported here are traceable to the project repository and its recorded experiment logs.

[**TODO:** Student Name 1 (Roll No.)]
[**TODO:** Student Name 2 (Roll No.)]

Acknowledgement

[**TODO:** Acknowledgement. Thank supervisor(s), department, Kaggle for GPU time, the DIV2K and KAIR authors. Half a page, plain and specific.]

Abstract

[**TODO:** Write last, about 250 words, one paragraph. Cover: the problem (composite archival degradation under a 5M parameter / 10 GFLOP budget); the method (fixed radial frequency split, shared 7x7/5x5/3x3 experts, spatial top-2 gate); the result (27.24 dB on 100 DIV2K validation images with 97,088 parameters against 27.83 dB for DnCNN at 558,403, slower on a T4 despite fewer FLOPs); the routing finding (gate tracks noise level, r = +0.81; overriding it costs 0.37 dB); and the status of the matched routing ablation.]

Keywords: image restoration, frequency decomposition, mixture of experts, spatial routing, lightweight networks

Contents

Certificate of Examination i

Supervisors' Certificate ii

Declaration of Originality iv

Acknowledgement v

Abstract vi

List of Figures ix

List of Tables x

List of Abbreviations xi

1 Introduction 1

1.1 Motivation . . . 1
1.2 Problem Statement . . . 1
1.3 Research Hypothesis and Objectives . . . 1
1.4 Contributions . . . 1
1.5 Organisation of the Report . . . 1

2 Background and Related Work 2

2.1 Image Restoration with CNNs . . . 2
2.2 Frequency-Domain Restoration . . . 2
2.3 Mixture of Experts and Gating . . . 2
2.4 Evaluation Metrics . . . 2
2.5 Research Gap . . . 2

3 Proposed Method: FCSG-Net 3

3.1 Degradation Model . . . 3
3.2 Architecture Overview . . . 3
3.3 Frequency Decomposition . . . 3

3.4 Shared Kernel-Size Experts . . . 3
3.5 Spatial Top-2 Gate . . . 5
3.6 Fusion and Residual Refinement . . . 5
3.7 Training Objective . . . 5
3.8 Model Size and Complexity . . . 5

4 Experimental Setup and Verification **7**
4.1 Dataset and Data Pipeline . . . 7
4.2 Baselines . . . 7
4.3 Training Protocol . . . 8
4.4 Evaluation Protocol . . . 8
4.5 Implementation Verification . . . 8

5 Restoration Results and Computational Cost **9**
5.1 Quantitative Comparison . . . 9
5.2 Per-Image Analysis . . . 10
5.3 Qualitative Results . . . 10
5.4 Training Convergence . . . 10
5.5 Latency, Memory and Tiling . . . 10

6 Routing Analysis and Ablation **11**
6.1 Spatial Routing Statistics . . . 11
6.2 Routing Versus Degradation Parameters . . . 11
6.3 Inference-Time Interventions . . . 12
6.4 Trained Routing Ablation . . . 12
6.5 Discussion . . . 12

7 Conclusion and Future Work **13**
7.1 Summary of Findings . . . 13
7.2 Limitations . . . 13
7.3 Future Work . . . 13

References **14**

A Reproducibility Record **15**
A.1 Code and Environment . . . 15
A.2 Source Bundles and Checkpoints . . . 15
A.3 How to Reproduce . . . 15

B Per-Image Results **16**

List of Figures

3.1 FCSG-Net data flow. Expert weights are shared across the three bands. 4
3.2 Soft radial band masks. 4
3.3 Spatial gate and top-2 expert selection. 5

4.1 Training data pipeline. 7

5.1 Quality against model size and compute. 9

6.1 Per-band expert weights for DIV2K image 0834. Rows: low, mid, high band. 11

List of Tables

2.1 Positioning of FCSG-Net against closely related methods. 2

3.1 Composite degradation pipeline. Parameters are sampled per image. . 3
3.2 FCSG-Net parameter breakdown. 6

4.1 Training hyperparameters, identical for every method. 8
4.2 Pre-training verification gates (Kaggle T4). 8

5.1 Composite restoration on 100 DIV2K validation images after 200k steps. Input PSNR is 20.992 dB for every method. 9

6.1 PSNR change when the trained gate is overridden at inference (10 images). 12
6.2 Learned top-2 routing against fixed uniform mixing, matched training. [**TODO:** fill from final runs] 12

A.1 SHA-256 identifiers (first 16 hex digits) of evaluated artefacts. 15

List of Abbreviations

AWGN	Additive White Gaussian Noise
CNN	Convolutional Neural Network
DIV2K	DIVerse 2K resolution image dataset
FFT	Fast Fourier Transform
FLOP	Floating-Point Operation
JPEG	Joint Photographic Experts Group (compression)
LPIPS	Learned Perceptual Image Patch Similarity
MoE	Mixture of Experts
PSNR	Peak Signal-to-Noise Ratio
SE	Squeeze-and-Excitation
SSIM	Structural Similarity Index

Chapter 1

Introduction

1.1 Motivation

[**TODO:** write 2-3 paragraphs]

1.2 Problem Statement

[**TODO:** write]

1.3 Research Hypothesis and Objectives

[**TODO:** write]

1.4 Contributions

[**TODO:** write]

1.5 Organisation of the Report

[**TODO:** write last]

Chapter 2

Background and Related Work

2.1 Image Restoration with CNNs

[**TODO:** adapt from notes/related.md "Foundations"]

2.2 Frequency-Domain Restoration

[**TODO:** adapt]

2.3 Mixture of Experts and Gating

[**TODO:** adapt]

2.4 Evaluation Metrics

[**TODO:** write]

2.5 Research Gap

Table 2.1: Positioning of FCSG-Net against closely related methods.

Method	Freq. split	Learned routing	Spatial	Notes
DnCNN [1]	–	–	–	residual CNN
FFDNet [2]	–	–	–	noise map input
MWCNN [3]	wavelet	–	–	joint subbands
SFNet [4]	learned	[**TODO:** check]	[**TODO:** check]	closest prior
Sparse MoE [5]	–	yes	–	token routing
FCSG-Net (ours)	radial FFT	top-2	per pixel	97k params

Chapter 3

Proposed Method: FCSG-Net

3.1 Degradation Model

[**TODO:** write]

Table 3.1: Composite degradation pipeline. Parameters are sampled per image.

Stage	Parameter	Distribution
Gaussian blur	σ_{blur}	$\mathcal{U}(0.5, 2.0)$ px
Down/up resampling	scale s	$\mathcal{U}(1.0, 2.0)$, bicubic
Additive noise	σ_n	$\mathcal{U}(5, 50)/255$, clipped
JPEG compression	quality q	integer $\mathcal{U}\{30, \ldots, 95\}$, 4:4:4

3.2 Architecture Overview

$$y_b(u, v) = \sum_e w_{b,e}(u, v)\, E_e(x_b)(u, v) \tag{3.1}$$

$$\hat{f} = x + \text{Refine}\big(\text{SEFusion}(y_L, y_M, y_H)\big) \tag{3.2}$$

3.3 Frequency Decomposition

[**TODO:** write; include mask equation with r1, r2, transition width]

3.4 Shared Kernel-Size Experts

[**TODO:** write]

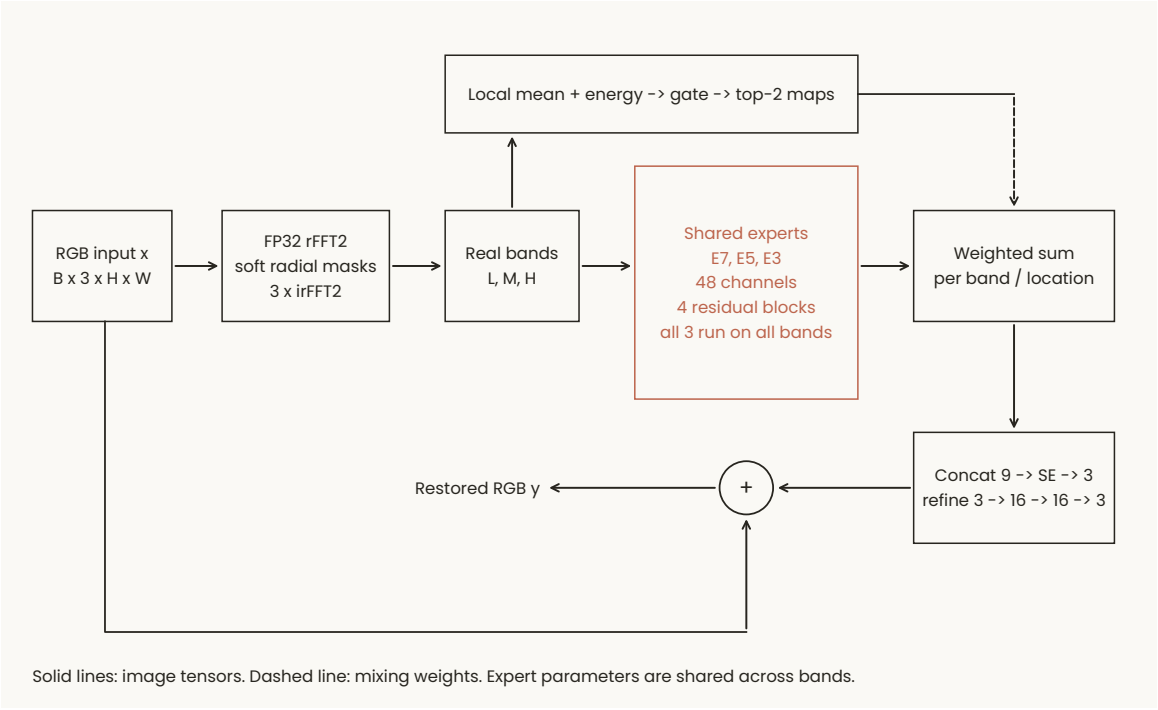


Figure 3.1: FCSG-Net data flow. Expert weights are shared across the three bands.

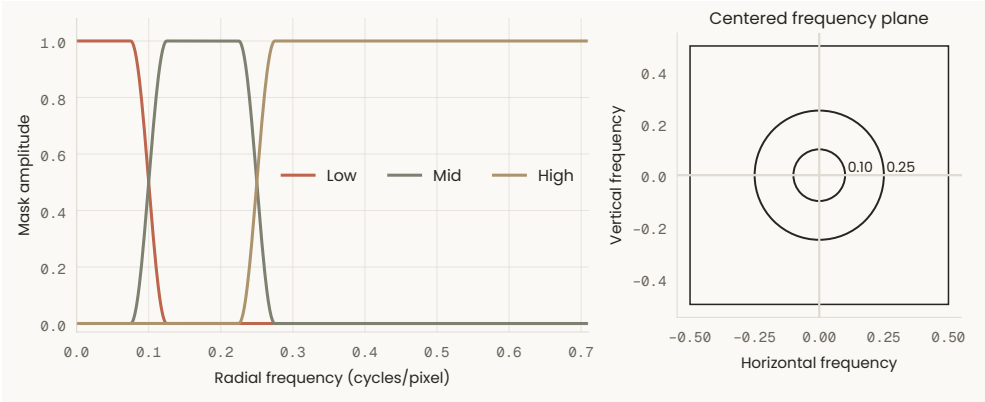


Figure 3.2: Soft radial band masks.

3.5 Spatial Top-2 Gate

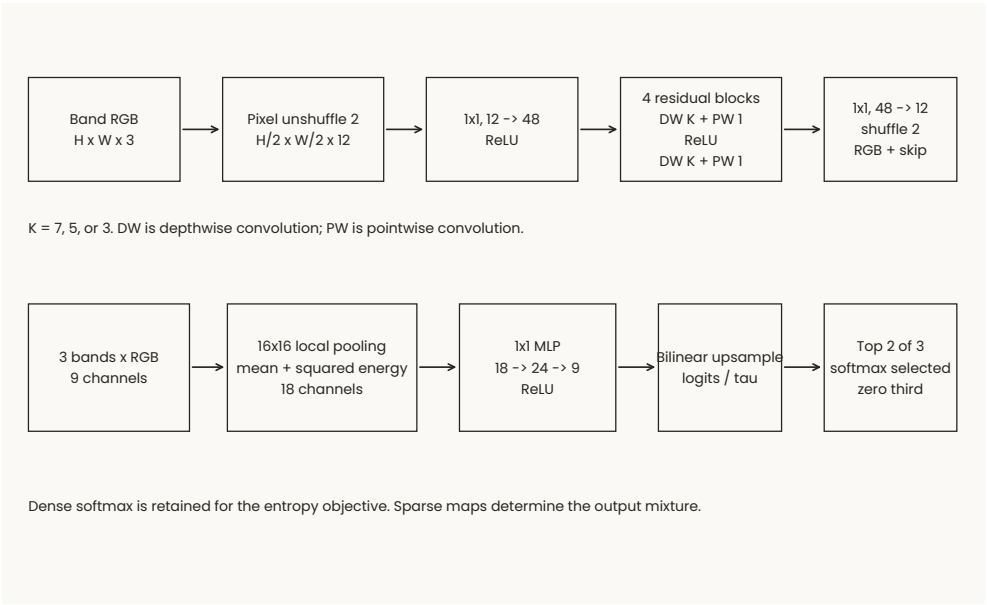


Figure 3.3: Spatial gate and top-2 expert selection.

[**TODO:** write; include the top-k softmax equation]

3.6 Fusion and Residual Refinement

[**TODO:** SE fusion 9-¿3 ch, then 3 conv refine 3-16-16-3, residual output]

3.7 Training Objective

$$\mathcal{L} = \mathcal{L}_{\text{char}} + 0.05\,\mathcal{L}_{\text{freq}} + \lambda_{\text{ent}}(t)\,\mathcal{L}_{\text{ent}} \tag{3.3}$$

[**TODO:** define each term]

3.8 Model Size and Complexity

Table 3.2: FCSG-Net parameter breakdown.

Module	Parameters
Shared 7×7 expert	39,228
Shared 5×5 expert	30,012
Shared 3×3 expert	23,868
Spatial gate ($18 \rightarrow 24 \rightarrow 9$)	681
SE fusion and $9 \rightarrow 3$ projection	96
Residual refinement	3,203
Total	97,088

Chapter 4

Experimental Setup and Verification

4.1 Dataset and Data Pipeline

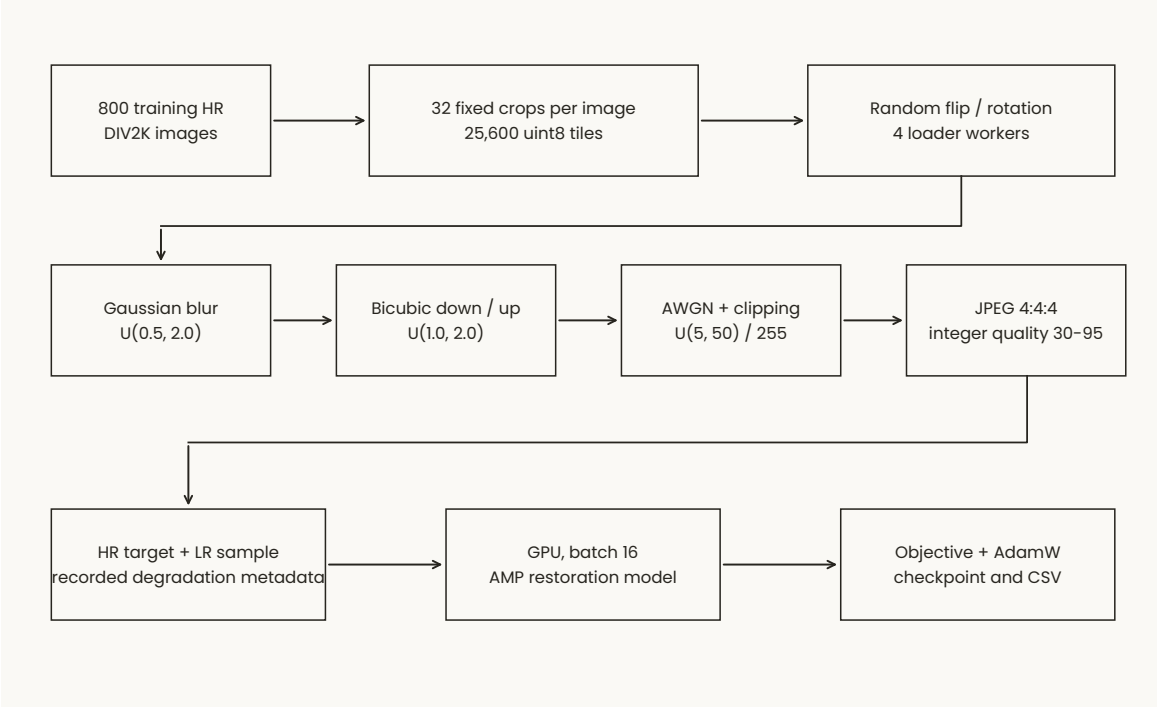


Figure 4.1: Training data pipeline.

[**TODO:** write]

4.2 Baselines

[**TODO:** write]

4.3 Training Protocol

Table 4.1: Training hyperparameters, identical for every method.

[**TODO:** fill from configs/*.toml]

[**TODO:** write; reuse schedules.pdf for learning-rate and entropy-weight schedules]

4.4 Evaluation Protocol

[**TODO:** write]

4.5 Implementation Verification

Table 4.2: Pre-training verification gates (Kaggle T4).

Check	Criterion	Measured
FFDNet reproduction (CBSD68, σ=25)	within 0.5 dB of 31.21	31.220 dB
Band reconstruction	$x_L + x_M + x_H = x$	max error 6.0×10^{-7}
Parameter budget	< 5M	97,088
Compute budget	< 10 GFLOPs @256	9.39
Memory	< 4 GB	0.60 GB peak reserved
Single-image overfit	MSE ratio < 0.25	0.19
Checkpoint resume	identical outputs	passed

Chapter 5

Restoration Results and Computational Cost

5.1 Quantitative Comparison

Table 5.1: Composite restoration on 100 DIV2K validation images after 200k steps. Input PSNR is 20.992 dB for every method.

Method	PSNR↑	Gain	SSIM↑	LPIPS↓	Params	GFLOPs
DnCNN	27.831	6.838	0.7795	0.3242	558,403	73.43
FFDNet (blind)	28.190	7.197	0.7899	0.3176	852,108	27.89
FFDNet (oracle)†	28.240	7.247	0.7920	0.3162	852,108	27.89
FCSG-Net	27.243	6.250	0.7529	0.3961	97,088	9.39

†Given the true noise σ; an upper bound, not a fair competitor.

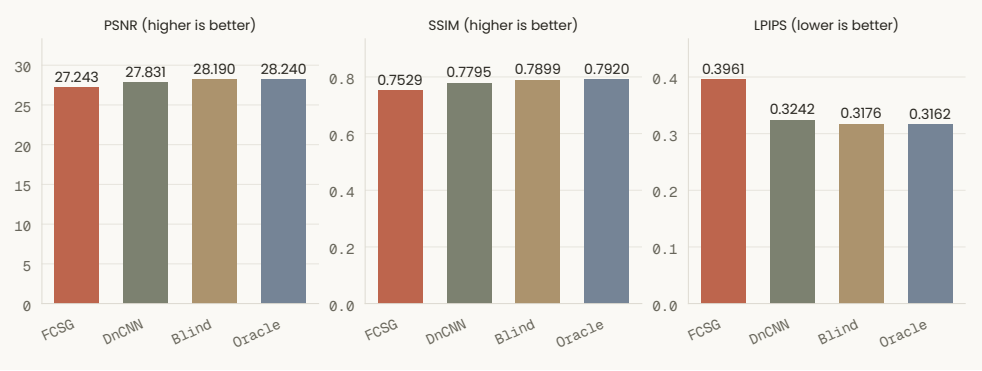


Figure 5.1: Quality against model size and compute.

[**TODO:** write: what the gap is, 82.6% fewer params and 87.2% fewer GFLOPs than DnCNN]

5.2 Per-Image Analysis

[**TODO:** write]

5.3 Qualitative Results

[**TODO:** assemble subfigure panel]

5.4 Training Convergence

[**TODO:** write]

5.5 Latency, Memory and Tiling

[**TODO:** write]

Chapter 6

Routing Analysis and Ablation

6.1 Spatial Routing Statistics

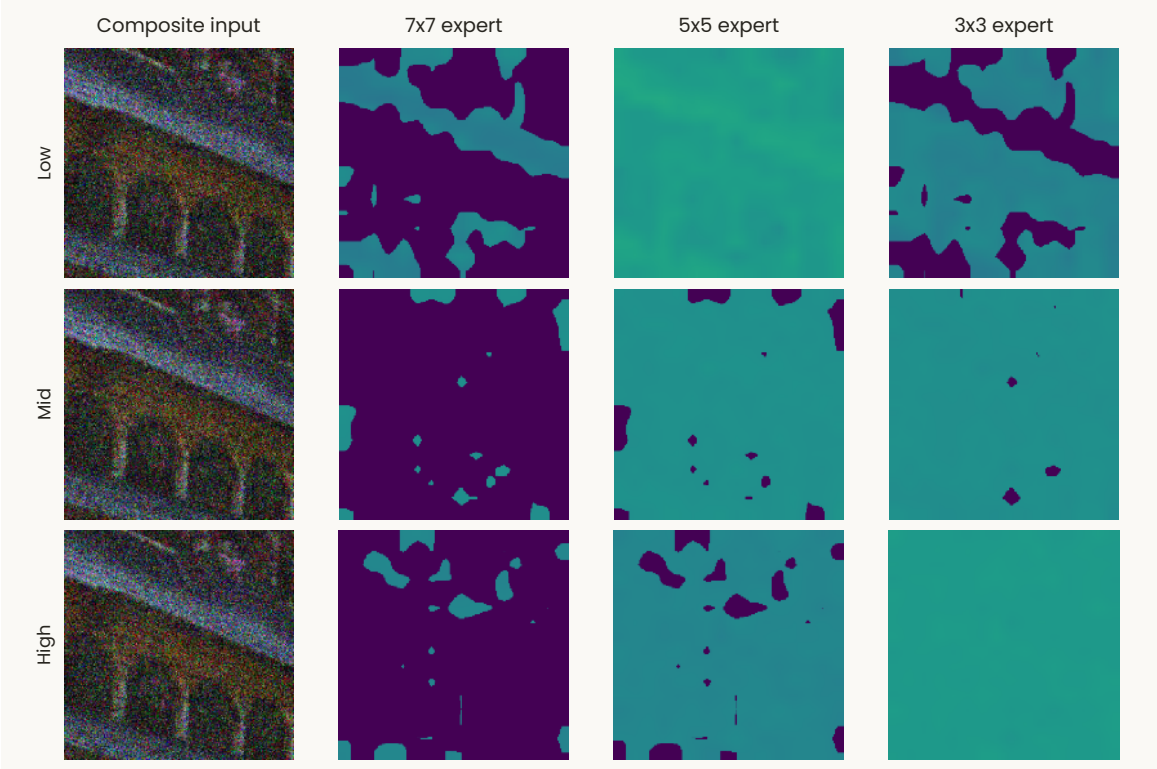


Figure 6.1: Per-band expert weights for DIV2K image 0834. Rows: low, mid, high band.

[**TODO:** write]

6.2 Routing Versus Degradation Parameters

[**TODO:** write, with the caveats above]

6.3 Inference-Time Interventions

Table 6.1: PSNR change when the trained gate is overridden at inference (10 images).

Intervention	Mean ΔPSNR	Min / Max
Equal weights on selected pair	−0.040 dB	−0.137 / +0.028
Uniform weights over all experts	−0.372 dB	−0.986 / +0.163

[**TODO:** write]

6.4 Trained Routing Ablation

Table 6.2: Learned top-2 routing against fixed uniform mixing, matched training. [**TODO:** fill from final runs]

Variant	PSNR	SSIM	LPIPS	Trainable params
Reference (learned top-2)	–	–	–	97,088
Uniform ($w = 1/3$)	–	–	–	96,407

[**TODO:** write once numbers exist]

6.5 Discussion

[**TODO:** write]

Chapter 7

Conclusion and Future Work

7.1 Summary of Findings

[**TODO:** write]

7.2 Limitations

[**TODO:** write as prose, not a dump of the table]

7.3 Future Work

[**TODO:** write]

References

[1] K. Zhang, W. Zuo, Y. Chen, D. Meng, and L. Zhang, "Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising," *IEEE Transactions on Image Processing*, vol. 26, no. 7, pp. 3142–3155, 2017.

[2] K. Zhang, W. Zuo, and L. Zhang, "FFDNet: Toward a fast and flexible solution for CNN-based image denoising," *IEEE Transactions on Image Processing*, vol. 27, no. 9, pp. 4608–4622, 2018.

[3] P. Liu, H. Zhang, K. Zhang, L. Lin, and W. Zuo, "Multi-level wavelet-CNN for image restoration," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2018.

[4] Y. Cui, Y. Tao, Z. Bing, W. Ren, X. Gao, X. Cao, K. Huang, and A. Knoll, "Selective frequency network for image restoration," in *International Conference on Learning Representations*, 2023.

[5] N. Shazeer, A. Mirhoseini, K. Maziarz, A. Davis, Q. Le, G. Hinton, and J. Dean, "Outrageously large neural networks: The sparsely-gated mixture-of-experts layer," in *International Conference on Learning Representations*, 2017.

Appendix A

Reproducibility Record

A.1 Code and Environment

[**TODO:** fill]

A.2 Source Bundles and Checkpoints

Table A.1: SHA-256 identifiers (first 16 hex digits) of evaluated artefacts.

Artefact	SHA-256 prefix
Phase 3 source bundle	`85d6e16f72ead013`
Diagnostics source bundle	`d473bc10b0ad0c1e`
Ablation source bundle	`4617fba4429c7b7f`
D2 paired-data manifest	`7ce7f310bdb4d2d5`
FCSG-Net checkpoint	`8db0bb0f3c251ddc`
DnCNN checkpoint	`afb0abfa022ede04`
FFDNet blind checkpoint	`981560c18cbea134`
FFDNet oracle checkpoint	`ef9560e8ed1ebae0`

A.3 How to Reproduce

[**TODO:** fill]

Appendix B

Per-Image Results

[**TODO:** generate table]